Bio303 - FINALTERM SUBJECTIVE MEGA FILE

ALL DATA FROM PAST PAPERS

GROUP: BIOTECH BRAINY BUNCH

ADMINS: MEHWISH MUGHAL & BELLA

CONTACT: mehwishshabirhussain07@gmail.com

Final Term Subjective Current + Past questions

1. Glycolysis?

Glycolysis is a cytoplasmic pathway which breaks down glucose into two three-carbon compounds and generates energy. Glucose is trapped by phosphorylation, with the help of the enzyme hexokinase. Adenosine triphosphate (ATP) is used in this reaction and the product, glucose-6-P, inhibits hexokinase.

2. Why pure water carries current?

Pure water contains electrical current because H_3O^+ migrates towards the cathode and OH^- towards anode.

3. First order reaction? **OR**

What is the first order reaction in terms of substrate?

For the uni molecular reaction S → P, the rate (or velocity) of the reaction, V—representing the amount of S that reacts per unit time—is expressed by a rate equation: V = k[S]. An equilibrium such as S ↔ P is described by an equilibrium constant, Keq, or simply K.

Under the standard conditions used to compare biochemical processes, an equilibrium constant is denoted K′eq (or K′). In this reaction, the rate depends only on the concentration of S. This is called a first-order reaction. he factor k is a proportionality constant that reflects the probability of reaction under a given set of conditions (pH, temperature, and so forth).

4. Source of fatty acids in cells?

To utilize acetate from fat for biosynthesis of carbohydrates, the glyoxylate cycle, whose initial reactions are identical to the TCA cycle, is used. This pathway thus allows cells to obtain energy from fat. The oxaloacetate can be converted into Phosphoenolpyruvate, which is the product of phosphoenol pyruvate carboxykinase, the first enzyme in gluconeogenesis.

5. Advantages of using triacylglycerol as storage source?

1. Fats are highly reduced hydrocarbons with a large energy of complete oxidation (38 kJ/g) more than twice that for the same weight of carbohydrate or protein.

2. Fats are chemically inert (triglycerides or neutral fats)

that make them especially suitable as storage fuels so they can be stored without fear of unfavorable reactions.

3. Fats are insoluble molecules that aggregate into droplets.

6. Function of coenzyme A in pyruvate dehydrogenase?

Coenzyme A (CoA, sometimes denoted CoA-SH, to emphasize the role of the OSH group)

Coenzyme A (CoA). A hydroxyl group of pantothenic acid
is joined to a modified ADP moiety by a phosphate ester bond, and its carboxyl group is attached to b-mercaptoethylamine in amide linkage.
The hydroxyl group at the 3’ position of the ADP moiety has a phosphoryl group not present in free ADP.

The —SH group forms a thioester with acetate in acetyl-coenzyme A (acetyl-CoA).

7. How water ionize and what happens to hydrogen ions? OR

Which ions are formed due to ionization of water.? and what happens to H+ ions?

DO NOT COPY

Water molecule have a slight tendency to undergo reversible ionization to yield a hydrogen ion (a proton) and a hydroxide ion, giving the equilibrium reaction. Hydrogen ion formed in water are immediately hydrated to hydronium ions (H_3O^+).

8. How malate synthase throw light on evolutionary history of glyoxylate?

In recent years, the detection of malate synthase (MS) and isocitrate lyase (ICL), key enzymes involved in the glyoxylate cycle, in some animal tissue has raised questions regarding the evolutionary relationship of enzymes in bacteria and animals and suggests that animals encode alternative enzymes of the cycle that differ in function from known MS and ICL in non-metazoan species. Isocitrate lyase and malate synthase are unique to the glyoxylate cycle. Notice that two acetyl groups (light red) enter the cycle and four carbons leave as succinate (blue). Glyoxylate is subsequently combined with acetyl-CoA to produce malate, catalyzed by malate synthase. Malate is also formed in parallel from succinate by the action of succinate dehydrogenase and fumarase.

9. Enzyme kinetics when enzyme concentration is less than substrate?

A numerically large (high) Km reflects a low affinity of enzyme for substrate because a high concentration of substrate is needed to half-saturate the enzyme.

10. Define pH? Derived its equation. Explain in terms of water pH at 25℃?

pH: The pH of a solution is defined as the logarithm to the base 10 of the reciprocal of the [H+], i. e, the negative logarithm of the [H+].

Equation:

Designates the H+ and OH- Concentrations

pH = log 1/[H+]

pH = -log[H+]

For each pH unit less than 7.0, the [H+] is increased tenfold.

for each pH unit above 7.0, it is decreased tenfold.

Water pH at 25℃:

The pH of water at 25°C, in which H+ and OH– ions are present in equal numbers, is 7.0

pH = -log[1 x 10-7]= 7

The symbol p denotes "negative logarithm of"

DO NOT COPY

11. 1000ml of water is equal to 1000g. Explain scientific reason?

12. What are the disadvantages of using triacylglycerol as storage?

1. Fats must be emulsified before enzymes can digest them.
2. Fats are insoluble in the blood and must be carried in the

blood as protein complexes.

13. What is non-competitive inhibitor. How it is different from competitive inhibitor?

Non-Competitive Inhibitors:

Noncompetitive inhibitors do not bind at the active site. They bind either free enzyme at a site other than active site or the ES complex.

Competitive Inhibitors:

This type of inhibition occurs when the inhibitor binds reversibly to the same site that the substrate would normally occupy i.e active site and, therefore, competes with the substrate for that site.

14. Define inhibitors and two classes of inhibitors?

Inhibitors:

Any substance that can diminish the velocity of an enzyme- catalyzed reaction is called an inhibitor.

Classes of Inhibitors:

Two broad classes of enzyme inhibitors:

1. Reversible
2. Irreversible.

15. Describe the oxidative pathway of pentose sugar pathway?

16. Name the 3 enzymes involved in non-reversible steps of Glycolysis?

DO NOT COPY

17. Describe the enzyme kinetics when substrate is in larger amount than the enzyme?

The effect of a competitive inhibitor is reversed by increasing [S].

At a sufficiently high substrate concentration, the reaction velocity reaches the Vmax as observed in the absence of inhibitor. When [S] far exceeds [I], the probability that an inhibitor molecule will bind to the enzyme is minimized and the reaction exhibits a normal Vmax. Therefore, a competitive inhibitor does not decrease Vmax.

18. Describe the function of isocitrate dehydrogenase?

1. Oxidation of Isocitrate to a-Ketoglutarate and CO2.

2. socitrate dehydrogenase catalyzes oxidative decarboxylation of isocitrate to form a-ketoglutarate

3. Mn2+ in the active site interacts with the carbonyl group of the intermediate oxalosuccinate, which is formed transiently but does not leave the binding site until decarboxylation converts it to a-ketoglutarate.

4. Mn2+ also stabilizes the enol formed transiently by decarboxylation.

Oxidation of the alcohol to ketone involves the transfer of a hydride from the C-H of the alcohol to the nicotinamide cofactor.

19. Malate dehydrogenase?

Oxidation of Malate to Oxaloacetate Is the last reaction of the citric acid cycle. NAD-linked L-malate dehydrogenase catalyzes the oxidation of L-malate to oxaloacetate. In intact cells oxaloacetate is continually removed by the highly exergonic citrate synthase reaction. This keeps the concentration of oxaloacetate in the cell remains extremely low pulling the malate dehydrogenase reaction toward the formation of oxaloacetate.

20. Industrial fermentation, 3 main stages name?

Industrial fermentation: It is the intentional use of fermentation in manufacturing products useful to humans. In addition to the mass production of fermented foods and drinks, industrial fermentation has widespread applications in chemical industry.

Stages: Industrial fermentations include both upstream (USP) and downstream processing (DSP) stages. USP consists of three main areas:

1. The producing organism

2. The culture medium

DO NOT COPY

3. The fermentation process

21. Malate formation?

Glyoxylate is subsequently combined with acetyl-CoA to produce malate, catalyzed by malate synthase. Malate is also formed in parallel from succinate by the action of succinate dehydrogenase and fumarase.

22. Phosphorylation compounds in biological reaction?

Phosphorylation is the addition of a phosphoryl (PO_3) group to a molecule. In biological systems, this reaction is vital for the cellular storage and transfer of free energy using energy carrier molecules.

23. Describe co-ordinations of glyoxylate cycle and citric acid?

Coordination of the glyoxylate cycle and the citric acid cycle:

The citric acid cycle is a major catabolic pathway producing a considerable amount of energy for cells, whereas the glyoxylate cycle's main function is anabolic - to allow production of glucose from fatty acids in plants and bacteria.

24. Reduction equivalent?

In biochemistry, the term reducing equivalent refers to any of a number of chemical species which transfer the equivalent of one electron in redox reactions. A redox reaction results in a change of oxidation state of atoms or ions due to actual or formal transfer of electrons.

25. Competitive inhibition and effect on Vmax?

Competitive Inhibition:

This type of inhibition occurs when the inhibitor binds reversibly to the same site that the substrate would normally occupy i.e active site and, therefore, competes with the substrate for that site.

Effect on Vmax:

The effect of a competitive inhibitor is reversed by increasing [S]. At a sufficiently high substrate concentration, the reaction velocity reaches the Vmax as observed in the absence of inhibitor. When [S] far exceeds [I], the probability that an inhibitor molecule will bind to the enzyme is minimized and the reaction exhibits a normal Vmax.

DO NOT COPY

Therefore, a competitive inhibitor does not decrease Vmax.

26. Actual free energy, how it determines directions of reaction, value in forward reverse reaction?

Actual Free Energy:

The standard free energy change ($\Delta G^{o'}$) of a chemical reaction is the amount of energy released in the conversion of reactants to products under standard conditions.

Directions:

When the free energy value is less than zero, the reaction will shift to the right, while when the free energy value is greater than zero, the reaction shifts to the left. When the free energy is zero, the reaction will be in a state of equilibrium and does not go in any direction.

Value for forward reaction:

The standard free energy change ($\Delta G^{o'}$) of a chemical reaction is the amount of energy released in the conversion of reactants to products under standard conditions.

27. Pyruvate dehydrogenase?

Pyruvate dehydrogenase complex carries out the five consecutive reactions in the decarboxylation and dehydrogenation of pyruvate. The PDH complex contains three enzyme.

1. Pyruvate dehydrogenase (E1)
2. Dihydrolipoyl transacetylase (E2)
3. Dihydrolipoyl dehydrogenase (E3)

The PDH complex isolated from mammals is about 50 nm in diameter--more than five times the size of an entire ribosome and big enough to be visualized with the electron microscope.

28. What is the activation energy reflect the rate of reaction?

The rate of reaction reflects this activation energy: higher activation energy corresponds to a slow reaction.

The function of cofactor in dehydrogenase coenzyme A?

The function of cofactor in dehydrogenase coenzyme A accepts acetyl group from Lipoamide.

DO NOT COPY

29. Describe mechanism of enzyme action and depict how activation energy affect reaction rate?

Mechanism of enzyme action:

Enzymes bind and chemically transform other molecules— they catalyze reactions. The molecules acted upon by enzymes are called reaction substrates rather than ligands. Enzymes are highly effective catalysts, commonly enhancing reaction rates by a factor of 10^5 to 10^{17}.

Effect on reaction rate:

The activation energies, G‡, for the

S → P and P → S reactions are indicated.

G′0 is the overall standard free- energy change in the direction S →P

Double dagger . Δ G‡.

30. What is half reaction?

A half reaction is either the oxidation or reduction reaction component of a redox reaction. A half reaction is obtained by considering the change in oxidation state of individual substance involved in the redox reaction.

31. Define activation energy?

Activation energy, in chemistry, is the minimum amount of energy that is required to activate atoms or molecules to a condition in which they can undergo chemical transformation or physical transport.

OR

The activation energy is the difference between the energy levels of the ground state and the transitional state.

32. Bioenergetics?

Bioenergetics is the branch of chemistry that focuses on how cells transform energy, often by producing, storing or consuming adenosine triphosphate (ATP). Bioenergetics processes, such as cellular respiration or photosynthesis, are essential to most aspects of cellular metabolism, therefore to life itself.

DO NOT COPY

33. Ionization?

Ionization, in chemistry and physics, any process by which electrically neutral atoms or molecules are converted to electrically charged atoms or molecules (ions) through gaining or losing electrons.

34. Define hyperchromicity of denaturation of DNA?

Hyperchromicity:

Hyperchromicity is the increase of absorbance of a material. The most famous example is the hyperchromicity of DNA that occurs when the DNA duplex is denatured.

Hyperchromicity in DNA denaturation:

When DNA in solution is heated above its melting temperature (usually more than 80 °C), the double-stranded DNA unwinds to form single-stranded DNA. The bases become unstacked and can thus absorb more light.

35. Define upstream processing?

Upstream processing includes formulation of the fermentation medium, sterilization of air, fermentation medium and the fermenter inoculum preparation and inoculation of the medium. The fermentation medium should contain any energy source, a carbon source, a nitrogen source and micro-nutrients required for the growth of the micro-organisms along with water and oxygen, if necessary.

36. Explain binding site of ligand?

A ligand may be any kind of molecule, including another protein.

1. A ligand binds at on the protein called the binding site.

2. The binding site complementary to the ligand in size, shape, charge and hydrophobic or hydrophilic character.

37. What are ligands?

In biochemistry and pharmacology, a ligand (from the Latin word, Ligandum meaning binding) is a substance that is usually a small molecule, that forms a complex with a biomolecule to serve a biological purpose. In a narrow sense, it is a single trigging molecule, binding to a site on a

DO NOT COPY

target protein. A molecule bound reversibly by a protein is called a ligand. Ligands include substrate inhibitors, activators and neurotransmitters.

38. What is ketoacidosis and how is it occur?

Diabetic ketoacidosis (DKA) is an acute, severe, and potentially life-threatening phenomenon that is caused by severe hyperglycemia, in which the blood glucose levels exceed 250 mg/dL. There is excess production of ketoacids due to a lack of insulin. Insulin deficiency can be absolute or relative.

39. Internal loop in RNA?

Internal-loops (also termed interior loops) in RNA are found where the double stranded RNA separates due to no Watson-Crick-Franklin base pairing between the nucleotides. Internal-loops differ from Stem-loops as they occur in middle of a stretch of double stranded RNA.

40. Name 2 molecules produced by Propioni bacterium?

Propioni bacterium is slow-growing, non spore-forming, Gram-positive, mesophilic, and tends toward anaerobiosis. It can be rod-shaped or branched, pleomorphic, and can occur singularly, in pairs, or in groups. It generally produces lactic acid, propionic acid, acetic acid, and CO_2 from glucose.

41. How ribonucleotides held together?

Ribonucleotides are linked together to form RNA strands via phosphodiester bonds. The 5'-phosphate group of one nucleotide is linked to the 3'-hydroxyl group of the next nucleotide, creating a backbone of alternating phosphate and pentose residues.

42. How many molecules of NADH are used during glycolysis?

Glycolysis starts with one molecule of glucose and ends with two pyruvate (pyruvic acid) molecules, a total of four ATP molecules, and two molecules of NADH.

43. What are isozymes?

Two or more enzymes that catalyze the same reaction but are encoded by different genes are called isozymes.

DO NOT COPY

44. What are thioalcohols?

Thioalcohols (thiols):

The oxygen atom of an alcohol is replaced with a sulfur atom.

45. Write equation of gluconeogenesis?

Glucose 6-phosphatase + H_2O --------- Glucose + Pi 36.

46. What are the components of eukaryotic ribosomes. Write about 60S subunits?

The mammalian ribosome contains two major nucleoprotein subunits including a large one with 60S and a smaller one with 40S. The 60S contains a 5S rRNA, 5.8S rRNA, a 28S rRNA, more than 50 polypeptides. The 40S subunit is smaller and contain a single 18S rRNA and approximately 30 distinct polypeptides chains.

47. Give names of some industrial fermentation and the products they form in industry?

Yogurt, is produced when the bacterium Lactobacillus bulgaricus ferments the carbohydrates in milk, producing lactic acid, the resulting drop in pH causes the milk proteins to precipitate, producing the thick texture and sour taste of unsweetened yogurt. Cheese is produced by using Propionibacterium Freudenreichii, which ferments milk to produce propionic acid and CO2. The propionic acid precipitates milk proteins, and bubbles of CO2 cause the holes.

48. Write equation of gluconeogenesis?

2 pyruvate + 4 ATP + 2 GTP + 2 NADH → Glucose + 4 GDP + 2 NAD + 6 inorganic phosphates (Pi).

Name two phases of pentose phosphate pathways?

There are two phases of pentose phosphate pathways that includes the oxidative phase and the non-oxidative phase.

49. Standard reduction potential?

When two conjugate redox pairs are together in solution, electron transfer from the electron donor of one pair to the electron acceptors of the other may proceed spontaneously. The tendency for such a reaction depends on the relative affinity of the electron acceptors of each redox pair for electrons. Measure of this affinity (in volts) is called the standard reduction potential.

DO NOT COPY

50. Which form of pyruvate can immediately tautomerize? What does it form?

In this substrate level phosphorylation, the product pyruvate first appears in its enol form, then tautomerize rapidly and non-enzymatically to its keto form which predominates at pH 7 form.

51. Note worth chemical transformation during glycolysis?

Degradation of carbon skeleton of glucose to yield pyruvate.

Phosphorylation of ADP to ATP by compounds with high phosphoryl group transfer potential, formed during glycolysis

Transfer of a hydride ion to NAD-1, forming NADH.

52. Define gluconeogenesis?

Glucose produced by gluconeogenesis in the liver or kidney or ingested in the diet is delivered to these other tissues, including brain and muscle, through the blood stream.

53. Name of the enzyme require for third bypass of gluconeogenesis?

Glucose 6-phosphate

54. Define Gibbs free energy?

Gibbs free energy, G expresses the amount of energy capable of doing work during a reaction at constant temperature and pressure. When a reaction proceeds with the release of free energy (that is when the system changes so as to possess less free energy), the free energy change, G, has a negative value and the reaction is said to be exergonic. In endergonic, the system gains free energy and G is positive.

55. Why sterilization is essential in fermentation?

Sterilizing the feed solution is essential because the media cannot contain the foreign microbes as this could severely hinder the growth of the production microbe. Its most popular method is heat sterilization of the feed solution.

56. In which conditions renaturation of DNA take place?

DO NOT COPY

Under appropriate conditions such as temperature and salt concentrations, separated strands of DNA will renature or reassociate and form the double helix by the process called renaturation (or annealing).

57. Write 6 types of Fermenters?

1. Simple fermenter (batch and continuous)
2. Fed batch fermenter
3. Air-lift or bubble fermenter
4. Cyclone column fermenter
5. Tower fermenter
6. Fluidized bed bioreactors
7. Packed bed bioreactor
8. Photobioreactor

58. Glycolysis form pyruvate via 3 other courts, name them and write on oxidative state?

The oxidative state, sometimes referred to as oxidation number, describes degree of oxidation (loss of electrons) of an atom in a chemical compound. The term oxidative was first used by Antoine Lavoisier to signify reaction of a substance with oxygen. Much later, it was realized that the substance, upon being oxidized, loses electrons, and the meaning was extended to include other reactions in which electrons are lost, regardless of whether oxygen was involved.

59. The acceptor arm of tRNA?

Arms or loops of tRNA:

All tRNA molecules contain 4 main arms or loops. Acceptor arm: This is made up of unpaired sequence of cytosine-cytosine-adenine (CCA) at the 3’ end. The 3’OH group of adenine binds with the carboxylic group of a specific amino acid and carries it to ribosomes for protein synthesis.

60. Three major fats of animals and vascular plants?

In animal and vascular plants, glucose has three major fats. It may be stored (as a polysaccharide or as sucrose). Oxidized to three carbon compounds (pyruvate) via glycolysis to provide ATP and the metabolic intermediates. Oxidized via the pentose phosphate (phosphor-gluconate)

DO NOT COPY

pathway to yield ribose. Phosphate for nucleic acid synthesis and NADPH for reductive biosynthetic processes.

61. What is Hyperchromicity?

Concomitant with this denaturation of the DNA molecule is an increase in the optical absorbance of the purine and pyrimidine bases----- a phenomenon referred to as hyperchromicity of denaturation of DNA.

62. Write pyruvate three metabolism pathways?

Pyruvate is the end product of second phase of glycolysis. For each glycolysis molecule, two ATP are consumed in the preparatory phase and four ATP are produced in the payoff phase. This gives a net yield of two ATP per molecule of glucose converted to pyruvate. Each phosphoryl group, represented here as P, has two negative charges.

63. What do u know about snRNA?

Small nuclear RNA (snRNA) is large number of small stable RNA species found in eukaryotic cells. Most of them are complexed with proteins to form ribonucleoproteins. They are disturbed in the nucleus, in the cytoplasm or in both. They are significantly involved in rRNA and mRNA processing and gene regulations.

64. E2 function?

E2 is the point of connection for the prosthetic group lipoate (blue), attached through an amide bond to the amino group of a Lys residue ...reaches outward to touch the active sites of E1 molecules (yellow) arranged on the E2 core.

The domains of E2 are separated by linkers, sequences of 20 to 30 amino acid residues, rich in Ala and Pro and interspersed with charged residues; these linkers tend to assume their extended forms, holding the three domains apart.

65. Write brief details relevant to pH? **OR**

Define pH? Equation of pH? Ph of water at 25 Celsius?

pH: The pH of a solution is defined as the logarithm to the base 10 of the reciprocal of the [H+], i. e, the negative logarithm of the [H+].

DO NOT COPY

Equation:

Designates the H+ and OH- Concentrations

pH = log 1/[H+]

pH = -log[H+]

For each pH unit less than 7.0, the [H+] is increased tenfold.

for each pH unit above 7.0, it is decreased tenfold.

Water pH at 25℃:

The pH of water at 25°C, in which H+ and OH– ions are present in equal numbers, is 7.0

pH = -log[1 x 10-7]= 7

The symbol p denotes "negative logarithm of".

66. Difference between oxidation & reduction?

Oxidation	Reduction
1. Oxidation is the loss of electrons from an atom, molecule or an ion.	**1.** Reduction is the gain of electrons from an atom, molecule or an ion.
2. Oxidation state increases	**2.** Reduction state decreases
3. Release electrons to the surroundings	**3.** Obtain electrons from the surroundings
4. Causes the increase of a positive charge of a chemical species	**4.** Cause the increase of a negative charge of a chemical species
5. Occurs in reducing agents	**5.** Occurs in oxidizing agents

67. What is Inhibitor and also write about its types? OR

What is enzyme inhibitor, its types and explain them?

Inhibitor:

Enzyme inhibitors are molecular agents that interfere with catalysis, slowing or halting enzymatic reactions.

OR

DO NOT COPY

Any substance that can diminish the velocity of an enzyme-catalyzed reaction is called an inhibitor.

Explanation:

Enzymes catalyze virtually all cellular processes, so it should not be surprising that enzyme inhibitors are among the most important pharmaceutical agents known.

For example, aspirin inhibits the enzyme (cyclooxygenase) that catalyzes the first step in the synthesis of prostaglandins, compounds involved in many processes, including some that produce pain.

Classification:

Inhibitors can be classified on the basis of their site of action on the enzyme on whether they chemically modify the enzyme, or on the kinetic parameters they influence.

Two broad classes of enzyme inhibitors:
1. Reversible
2. Irreversible.

Irreversible Inhibitors: In general, irreversible inhibitors bind to enzymes through covalent bonds. The two most commonly encountered types of reversible inhibition are competitive and noncompetitive inhibitors.

Reversible Inhibitors: Reversible inhibitors typically bind to enzymes through noncovalent bonds, dilution of the enzyme–inhibitor complex results in dissociation of the reversibly bound inhibitor, and recovery of enzyme activity.

DO NOT COPY

68. Glyoxylate and citric acid cycle? **OR**

Coordination between glyoxylate and citric acid cycle? **OR**

Evolutionary history of citric acid cycle and glyoxylate?

Coordinated regulation of glyoxylate and citric acid cycles:

The central molecules between 2 cycles is isocitrate. The translocation of isocitrate between the citric acid cycle and the glyoxylate cycle is controlled by the activity of isocitrate dehydrogenase, which is regulated by reversible phosphorylation. A specific protein kinase phosphorylates and inactivates the dehydrogenase. This inactivation shunts isocitrate to the glyoxylate cycle, where it begins the synthetic route toward glucose.

A phosphoprotein phosphatase removes the phosphoryl group from isocitrate dehydrogenase, reactivating the enzyme. Reactivation of enzyme sends more isocitrate through the energy-yielding citric acid cycle. The regulatory protein kinase and phosphoprotein phosphatase are separate enzymatic activities of a single polypeptide.

Regulation Linkage:

Coordinated regulation of glyoxylate and citric acid cycles. Regulation of isocitrate dehydrogenase activity determines the partitioning of isocitrate between the glyoxylate and citric acid cycles. When the enzyme is inactivated by phosphorylation (by a specific protein kinase), isocitrate is directed into biosynthetic reactions via the glyoxylate cycle. When the enzyme is activated by dephosphorylation (by a specific phosphatase), isocitrate enters the citric acid cycle and ATP is produced.

Name 5 diseases resulting from ion channel defect?

Fermentation and its two types?

Fermentation:

Fermentation actually comes from Latin word “fervere” which means “to boil” So the word “Fermentation” is referred to physical state of boiling/bubbling

The bubbling appearance in fermentation is due to the production of CO2 bubbles caused by anaerobic catabolism of sugars.

Fermentation is a metabolic process that converts sugars to acids, gases, or alcohol The science of fermentation is known as ZYMOLOGY.

French microbiologist Louis Pasteur is often remembered for his insights into fermentation and its microbial causes.

It occurs in yeast and bacteria, and also in oxygen-starved muscle cells

In terms of BIOCHEMISTRY Fermentation relates to the generation of energy by catabolism of organic compounds.

Types of fermentation:

There are two types of fermentation on basis of processes, lactic acid fermentation and

alcoholic fermentation.

LACTIC ACID FERMENTATION

Homolactic Fermentation:

DO NOT COPY

HOMOLACTIC In homolactic fermentation, one glucose molecule is converted into two molecules of lactic acid

$C_6H_{12}O_6 \rightarrow 2\ CH_3CHOHCOOH$

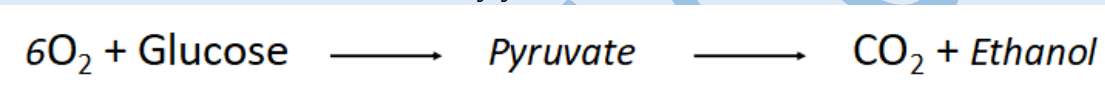
Heterolactic Fermentation:

In Heterolactic Fermentation, one glucose molecule is converted into one molecule of lactic acid, one molecule of ethanol and one molecule of carbondioxide

$C_6H_{12}O_6 \rightarrow CH_3CHOHCOOH + C_2H_5OH + CO_2$

ALCOHOL FERMENTATION

It is done by yeast and some kinds of bacteria.
Products: glucose undergoes fermentation to produce ethanol and CO2, instead of lactate. Substrate: glucose. Humans have long taken advantage of this process in making bread, beer, and wine. Alcohol fermentation is done by yeast and some kinds of bacteria.



Ethanol Is the Reduced Product in Ethanol Fermentation.

69. Write 10 steps of citric acid cycle with its enzyme names in each step?

Formation of citrate:

The first reaction of this cycle is the condensation of acetyl-CoA with oxaloacetate to form citrate, catalyzed by citrate synthase.

The methyl carbon of the acetyl group is joined to the carbonyl group (C-2) of oxaloacetate. Acetyl-CoA is a transient intermediate formed on the active site of the enzyme. It rapidly undergoes hydrolysis to release CoA and citrate.

The hydrolysis of this high energy thioester intermediate makes the forward reaction highly exergonic. The CoA liberated in this reaction is recycled to participate in the oxidative decarboxylation of another molecule of pyruvate by the PDH complex.

Isomerization of Citrate by Aconitase:

The enzyme aconitase (aconitate hydratase) catalyzed the reversible transformation of citrate to isocitrate.

Reaction intermediate of the tricarboxylic acid is formed known as cis-aconitate.

DO NOT COPY

Cis-aconitate does not dissociate form the active site.

Aconitase can promote the reversible addition of H2O to the double bound cis-aconitase in two different ways, one leading to citrate and the other to isocitrate.

Under physiological conditions in the cell, concentration of isocitrate is very little, the reaction is pulled to the right because isocitrate is rapidly consumed in the next step of this cycle.

The Isocitrate Dehydrogenase Reaction:

Oxidation of isocitrate to a-ketoglutarate and CO2 in the next step.

Isocitrate dehydrogenase catalyzes oxidative decarboxylation od isocitrate to form a-ketoglutarate.

Mn^{+2} in the active site interacts with the carbonyl group of the intermediate oxalosuccinate which is formed transiently but does not leave the binding site until decarboxylation converts it to a-ketoglutarate.

Mn^{+2} also stabilizes the enol formed transiently by decarboxylation.

Oxidation of a-Ketoglutarate:

The next step is another oxidative decarboxylation.

Oxidation of a-ketoglutarate to Succinyl-CoA and CO2.

a-Ketoglutarate is converted into succinyl-CoA and CO2 by the action of the a-Ketoglutarate dehydrogenase complex.

NAD^+ serves as electron acceptor and CoA as the carrier of the succinyl group.

The energy of oxidation of a a-ketoglutarate is conserved in the formation of the thioester bond of succinyl-CoA.

This reaction is somehow identical to the pyruvate dehydrogenase reaction and to the reaction sequence responsible for the breakdown of branched chain amino acids.

Conversion of Succinyl-CoA to succinate:

Succinyl-CoA like acetyl-CoA, has a thioester bond with a strongly negative standard free energy of hydrolysis (G=-36KJ/mol).

DO NOT COPY

Energy released in the breakage of this bond is used to derive the synthesis of a phosphoanhydride bond in either GTP or ATP, with a net G= - 2.9KJ/mol.

The enzyme that catalyzes this reversible reaction is called succinyl-CoA synthetase or succinate Thiokinase, names indicate the participation of a nucleotide triphosphate in the reaction.

Oxidation of Succinate to Fumarate:

Succinate dehydrogenase reaction

The succinate formed from succinyl-CoA is oxidized to fumarate by the flavoprotein succinate dehydrogenase.

In eukaryotes, succinate hydrogenase is tightly bound to the mitochondrial inner membrane, in bacteria, to the plasma membrane.

The enzyme contains three different iron-sulfur clusters and one molecule of covalently bound FAD.

Electron flow from succinate through these carriers to the final electron acceptors, O2 is coupled to the synthesis of about 1.5 ATP molecules per pair of electrons (respiration linked phosphorylation).

Hydration of Fumarate to Malate:

The reversible hydration of fumarate to L-malate is catalyzed by fumarase (fumarate hydratase).

The transition state in this reaction is a carbanion.

This enzyme is highly stereospecific, it catalyzes hydration of the trans double bond of fumarate but not the cis double bond of maleate (the cis isomer of fumarate).

In the reverse direction (from L-malate to fumarate), fumarase is equally stereospecific: D-malate is not a substrate.

Oxidation of Malate to Oxaloacetate:

Oxidation of malate to oxaloacetate is the last reaction of the citric acid cycle.

NAD linked L-malate dehydrogenase catalyzes the oxidation of L-malate to oxaloacetate.

In intact cells oxaloacetate is continually removed by the tightly exergonic citrate synthase reaction.

This keeps the concentration of oxaloacetate in the cell remains extremely low pulling the malate dehydrogenase reaction toward the formation of oxaloacetate.

70. TCA Cycle? **OR**

Brief description of citric acid cycle?

DO NOT COPY

Citric Acid Cycle:

1. Also called Tricarboxylic Acid Cycle (TCA) or Krebs Cycle (after its discoverer, Hans Krebs). Three names for the same thing.

2. A series of chemical reactions used by all aerobic organisms to generate energy through the OXIDATION of acetyl CoA ... derived from carbohydrates, fatty
acids and proteins into carbondioxide, water and chemical energy in the form of ATP.

3. Conversion of pyruvate to activated acetate.

4. Cellular respiration and intermediates for biosynthesis.

5. Conversion of acetate to carbohydrate precursors in the glyoxylate cycle.

71. Explain mechanism of enzyme action?

Enzymes bind and chemically transform other molecules-----they catalyze reactions.

1. The molecules acted upon by enzymes are called reaction substrates rather than ligands.

2. Enzymes are highly effective catalysts, commonly enhancing reaction rates by a factor of 10^5 to 10^{17}. The distinguishing feature of an enzyme catalyzed reaction is that it takes place within the confines of a pocket on the enzyme called the active site.

3. The surface of the active site is lined with amino acid residues with side chains that complements and bind the substrate and catalyze its chemical transformation

4. Often, the active site encloses a substrate, sequestering it completely from solution.

5. The enzyme substrate complex, whose existence is central to the action of enzyme.

When an enzyme binds to its substrate, we know it lower the activation energy of the reaction, allowing it to happen more quickly. The enzyme -substrate complex can also lower activation energy by bending substrate molecules in a way that facilities bond-breaking, helping to reach the transition state.

72. Oxidation of pyruvate and name three catabolic routes of glycolosis?

Oxidation of Pyruvate:

Pyruvate is oxidized, with loss of its carboxyl group as CO2 to yield the acetyl group of acetyl-coenzyme A, the acetyl group is then oxidized completely to CO2 by the citric acid cycle. The electron from this oxidation is passed to O2 through a chain of carriers in mitochondria, to form

DO NOT COPY

H_2O. The energy from the electron-transfer reactions drives the synthesis of ATP in mitochondria.

Catabolic routes of glycolysis:

1. Oxidation of pyruvate
2. Reduction of lactate
3. Ethanol (alcohol) fermentation

73. Function of phosphofructokinase?

1. Phosphofructokinase composed of PFKM subunits is involved in the sequence of events that breaks down glycogen to provide energy to muscle cells. Specifically, the enzyme converts a molecule called fructose-6-phosphate to a molecule called fructose 1,6-bisphosphate.

2. Phosphofructokinase, enzyme that is important in regulating the process of fermentation, by which one molecule of the simple sugar glucose is broken down to two molecules of pyruvic acid.

74. Write about structure and function of rRNA?

Ribosomal RNA (rRNA) is part of the ribosome, or protein builders, of the cell. Ribosomes are responsible for translation, or the process our cells use to make proteins. rRNA are responsible for reading the order of amino acids and linking amino acids together. They do this through a highly complex sequence.

75. Explain activation energy in detail?

Activation Energy:

The difference between the energy levels of the ground state and transition state is called activation energy, G‡.

Rate of Reaction:

The rate of reaction reflects this activation energy as a higher activation energy correspond to a slower reaction. Reaction rates can be increased by raising the temperature, thereby increasing the number of molecules with sufficient energy to overcome the energy barrier.

DO NOT COPY

Alternatively, the activation energy can be lowered by adding the catalyst. Catalysts enhance the reaction rates by lowering the activation energy. The role of enzyme is to accelerate the inter-conversion of S and P such as enzymes lower the energy of activation, Δ G‡, of a reaction. The enzyme is not used up in a process, so the equilibrium point is unaffected. The role of enzyme is to accelerate the inter-conversion of S and P.

76. Components of fermenter?

1. Formulation of media to be used in culturing the organisms during development of inoculum and in the production fermenter.

2. Sterilization of a medium, fermenter and ancillary equipment.

3. Production of an active, pure culture in sufficient quantity to inoculate the production vessels.

4. The growth of an organisms in the production fermenter under optimum conditions for product formation.

5. The extraction of a product and its purification.

6. Disposal of effluents produced during the process.

7. In the cytosol, oxaloacetate is converted into phosphoenolpyruvate by PEP, "Carboxy-kinase".

8. The conversion to lactate to pyruvate in the cytosol or the transfer of reducing equivalent from "mitochondria" to cytosol in form of malate.

9. When reaction proceeds with release of free energy (that is, when the system changes so as to possess less free energy), the free energy change G, has a "negative rule".

10. A phosphoryl group (-PO32-) can be transferred to an alcohol (forming a phosphate-ester), or to a carboxylic acid (mixed anhydride).

11. The German Biochemist Otto Warburg first observed in 1928 that tumors of nearly all types carry out glycolysis at a much higher rate than normal tissues, even when oxygen is available. This is known as "Warburg effect".

12. A few types of "Coenzymes and proteins" serves as Universal Electron Carriers.

13. About carbon, fermentation medium, commensals (commercial yeast) doubt, the pathway of glycolysis was explained by Otto Warburg and Hans Von Euler-Chelpin in "Yeast".

14. ATP consumed and converts into "ADP + Pi".

DO NOT COPY

15. Alcohol manufactured by fermentation.

16. Hybridization, 3D-arm as it contains the base "dihydrouridine".

77. Write 5 steps of glycolysis with writing the enzyme names?

Glycolysis is the metabolic process which serves as foundation for both aerobic and anaerobic cellular respiration. In glycolysis, glucose is converted into pyruvate. Glucose is the six membered ring molecule found in the blood and usually is a result of breakdown of carbohydrates into sugars. It enters the cells through specific transporter proteins that move if from outside the cells into cell cytosol. All the glycolytic enzymes are found in the cytosol. The overall reaction of glycolysis which occurs in cytoplasm is represented simply as:

$C_6\ H_{12}\ O_6$ + 2NAD + 2ADP + 2P → 2 pyruvic acid (CH_3(C=O)COOH + 2ATP +2 NADH + $2H^+$

The first step in the glycolysis is conversion of D-glucose into glucose-6-phosphate. The enzyme that catalyzes this reaction is hexokinase. The second reaction of glycolysis is rearrangement of glucose-6-phosphate (G6P) into fructose-6-phosphate (F6P) by glucose phosphate isomerase (phosphor-glucose isomerase). Phosphofructokinase, with magnesium as a cofactor, changes fructose-6-phosphate into fructose 1,6- bisphosphate. The enzyme aldolase splits the fructose 1,6-bisphosphate into two sugars that are isomers of each other. These two sugars are dihydroxyacetone phosphate (DHAP) and glyceraldehyde 3-phosphate (GAP). The enzyme triphosphate isomerase rapidly inter-converts the molecules dihydroxyacetone phosphate and glyceraldehyde 3-phosphate. Glyceraldehyde 3-phosphate is removed / used in further procedure of glycolysis.

78. Write steps of synthesis of phosphoenolpyruvate from pyruvate?

1. The first of the bypass reaction in gluconeogenesis is the conversion of pyruvate to phosphoenolpyruvate (PEP). This reaction cannot be occurred by simple reversal of pyruvate-kinase reaction of glycolysis which has a large, negative free energy change and is irreversible.

2. Instead, the phosphorylation of pyruvate is achieved by round about sequence of reactions that in eukaryotes requires enzymes in both cytosol and mitochondria. Pyruvate is first transported from the cytosol into mitochondria or is generated by transamination, in with alpha amino group is transferred from alanine (leaving pyruvate), to an alpha keto-carboxylic acid.

DO NOT COPY

3. In mitochondria, pyruvate is converted into oxaloacetate in a biotin (co-enzyme) requiring reaction catalyzed by pyruvate carboxylase, a mitochondrial enzyme.

Separate Whatsapp Groups for Girls and Boys

Special Internship Group with Senior Guideline

Zero Percentage Irrelevant Chatting in study groups

Follow this Link for Joining Study Group:

https://chat.whatsapp.com/D2yLdI6LFFpJuS2EKWYmbE

Remember Us in Your Prayers

DO NOT COPY